How do we write chemical reactions?

We use **word equations** to write down what happens in a **chemical reaction**. The equation shows the substances that are reacting **(reactants)** and what the reaction produces **(products)**.

reactants ⟶ products

If you heat copper carbonate, a chemical reaction occurs and copper oxide and carbon dioxide are produced. This can be written as the following word equation:

copper carbonate $\xrightarrow{\text{heat}}$ copper oxide + carbon dioxide

☞ **1** Complete the following equations by writing in the correct **product** after the arrow. The first one has been done for you.

a iron + oxygen ⟶ iron oxide

b copper + sulphur ⟶ _____

c carbon + oxygen ⟶ _____

d hydrogen + chlorine ⟶ _____

e aluminium + bromine ⟶ _____

2 Complete the following equations by writing in the missing **reactant**.

a potassium + _____ ⟶ potassium iodide

b iron + _____ ⟶ iron chloride

c _____ + oxygen ⟶ magnesium oxide

d _____ + bromine ⟶ silver bromide

3 Complete the following equations by filling in the missing **reactants** or **products**.

a _____ + _____ ⟶ calcium oxide

b _____ + _____ ⟶ aluminium chloride

c lead oxide + _____ ⟶ lead + carbon dioxide

d _____ + iodine ⟶ hydrogen _____

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